Software Requirements Specification (SRS)

Primap (Web Application)

Document Control

Document Information	
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Version History

Version	**Date**	**Description of Changes**
1.0	February 10th, 2026	Initial submission

1. Introduction

1.1 Purpose

This document specifies the requirements for **Primap**, a web-based application that supports citizen-science survey walks for reporting observations of primates in Singapore, with a focus on the critically endangered Raffles' Banded Langurs (RBLs). The application aims to replace the current spreadsheet-based workflow utilised by volunteers and administrators of the Raffles' Banded Langur Working Group (RBLWG) and streamline the management of volunteers and reporting of sightings during survey walks, while maintaining the necessary flexibility for real-world field survey conditions.

1.2 Scope

Stakeholders

- **Volunteers:** Primary users who require an interface that is easy to learn and use both during field walks and after walks, including support for offline entry and draft workflows.
- **Dr Ang / Raffles' Banded Langur Working Group:** Organisers who manage survey rounds and walk slots, monitor volunteer participation, and export data for conservation analysis. They require tools for bulk scheduling, reminders, and oversight.
- **Supporting Organisations (e.g., National Parks Board, National University of Singapore):** Organisations that may require compliance with ethical and regulatory obligations.

In Scope

The system shall support:

Volunteer Participation

- Volunteer self-registration with administrator approval.
- Survey walk sign-up and cancellation.
- Basic beginner guidance to help first-time volunteers.
- Volunteers viewing:
    - their own records,
    - their progress toward minimum required walks,
    - a map visualisation of their own observations (optional).

Observation Reporting

- Volunteers submitting observation reports for primates, including:

    - observation outcome (Sighted / Not Sighted),
    - walk completion status (Completed / Partially Completed / Aborted),
    - date, time, primate species, number of primates, optional demographics, and behaviour notes,
    - immutable submissions once finalised (no post-editing),
    - GPS-based location capture as mandatory data,
    - optional extraction of GPS metadata from uploaded images
    - offline draft creation with subsequent synchronisation,
    - media uploads (photos/videos), limited per observation.

Administrative Functions

- Access control:
    - create, edit, disable/ban volunteer accounts,
    - role-based access (admin vs volunteer).
- Admin dashboard for survey management:
    - survey round management (open/close rounds),
    - walk slot management (CRUD),
    - bulk creation of walk slots based on rules (e.g., day/time pattern for N weeks),
    - monitoring attendance, cancellations, and volunteer participation statistics.
- Notifications:
    - reminders before walks,
    - cancellation alerts to remaining slot members,
    - administrative announcements (optional).
- Data portability and preservation:
    - exporting/importing data compatible with spreadsheet workflows,
    - full dataset export for backup and scientific analysis.
- Offline operational support:
    - guidance for preparing the application for offline usage (e.g., caching before walks).

Out of Scope

The system shall not support:

- Real-time tracking, prediction, or broadcasting of primate locations during active survey periods.
- Enforced rigid survey routes or fixed sighting times.
- Advanced in-app training or certification programs.
- Advanced automatic scientific analysis or interpretation of survey data.
- Synchronised mirroring with Google Sheets as a primary system.
- Multiple interface languages (English only).
- Multiple browsers beyond the chosen supported browser (Chrome only).

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
RBL	Raffles’ Banded Langur
RBLWG	Raffles’ Banded Langur Working Group
Primate	In-scope observed animals including RBL, Long-tailed Macaque, and Dusky Langur
Survey Walk	A scheduled volunteer walk for observation
Round	A 24-week survey period opened by administrators
Slot	A scheduled survey walk instance with capacity up to three volunteers
Sighted	An observation outcome in which one or more primates are observed during a survey walk
Not Sighted	An observation outcome in which no primates are observed during a survey walk
Observation	A report submitted for a survey walk outcome
Draft	A saved, incomplete observation not yet submitted

Conventions

- **shall** indicates a mandatory requirement
- **should** indicates a recommended requirement
- **may** indicates an optional requirement

1.4 References

1. Ang, A. *et al.* (2020) *Citizen Science Program for Critically Endangered Primates: A Case Study from Singapore*. Available at: http://static1.1.sqspcdn.com/static/f/1200343/28485746/1638300722537/PC35_Ang_et_al_Citizen_science.pdf?token=mcGZFzVqOwiQB1d4cFSIpC7/UJ4%3D

2. Rigger, M. (2025) *Langur Survey with Wai Yee*, *YouTube*. Available at: https://www.youtube.com/watch?v=sl5x6ASDs0g

3. Rigger, M. (2026) *Workshop Andie + NUS*, *YouTube*. Available at: https://www.youtube.com/watch?v=lVctkMyS5uA

1.5 Overview

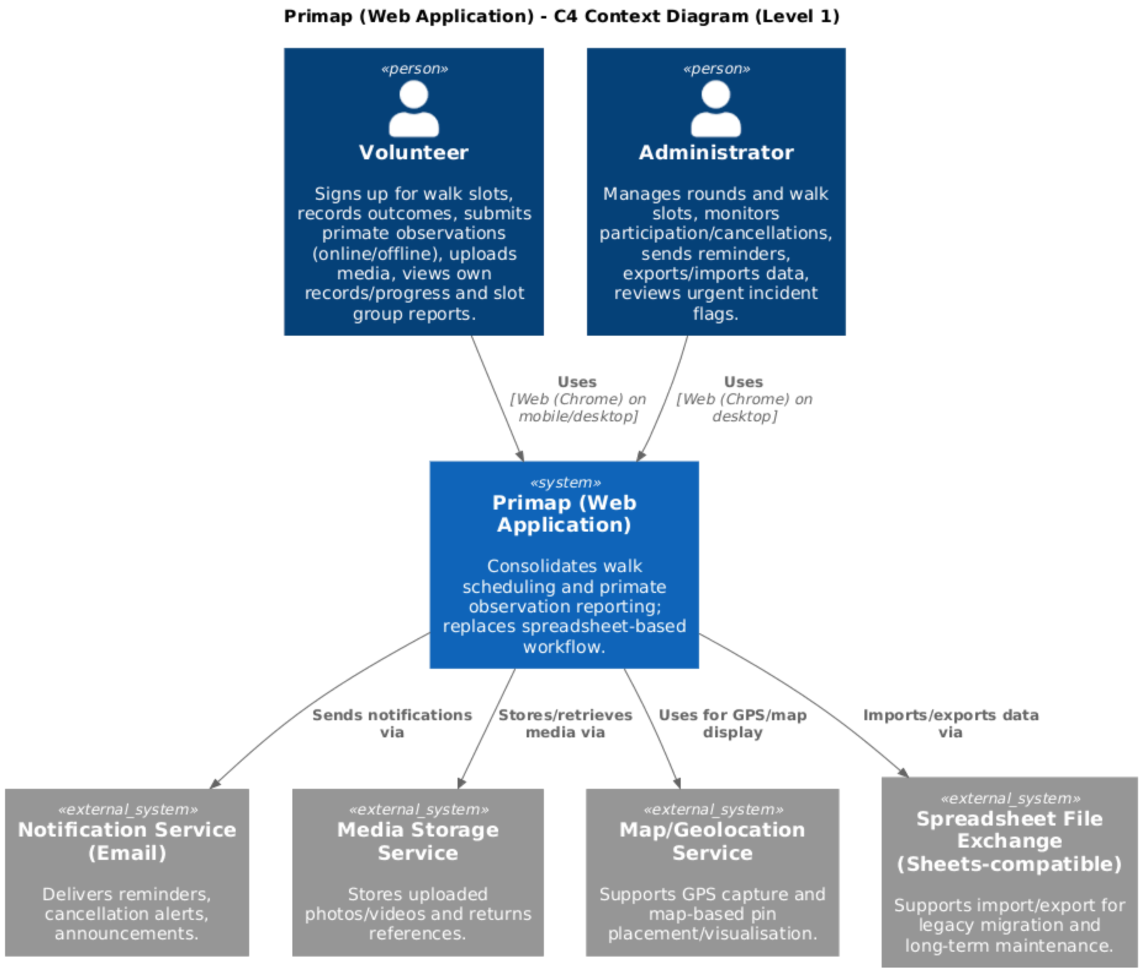
Section 2 presents the overall description of the system. It covers the product perspective and how Primap fits within its broader environment, the characteristics of its target users, the constraints governing its design and operation, and the assumptions and external dependencies the system relies on.

Section 3 defines the specific requirements of the system. It details the functional requirements from both volunteer and administrator perspectives, presents key use cases that describe how users interact with the system, specifies non-functional requirements covering usability, reliability, performance, and privacy, and outlines the data requirements necessary to support conservation reporting and administrative workflows.

2. Overall Description

2.1 Product Perspective

Primap is a standalone web application designed to replace the existing spreadsheet-based workflow used by the Raffles' Banded Langur Working Group. It integrates volunteer sign-ups, walk scheduling, and observation reporting into a single system. The system depends on external services to support authentication, persistent data storage, media storage, notification delivery, and map-based visualisation. A C4 Context diagram is used to illustrate users, Primap, and its external dependencies.



2.2 User Characteristics

Persona 1: Experienced Volunteer ("Eddie")

- **Age / Background:** 34, wildlife hobbyist, works in IT
- **Technical proficiency:** Moderate
- **Devices:** Mobile phone + laptop
- **Behaviors:**
    - Records photos on-site
    - Completes reports later on desktop
- **Needs:**
    - Fast data entry
    - Draft-saving and offline support
    - Minimal hassle

Persona 2: New Volunteer ("Neo")

- **Age / Background:** 19, undergraduate, first-time volunteer
- **Technical proficiency:** Moderate
- **Devices:** Mobile phone
- **Behaviors:**
    - Unsure about species identification
      Hesitant when reporting sightings
- **Needs:**
    - Beginner guidance
    - Simple to understand forms
    - Feedback and confirmation

Persona 3: Administrator ("Andie")

- **Age / Background:** 40s, conservation researcher
- **Technical proficiency:** Moderate to high
- **Devices:** Laptop (primary)
- **Behaviors:**
    - Manages rounds, walks, and volunteers
    - Exports data for research
- **Needs:**
    - Bulk scheduling tools
    - Participation analytics
    - Automated reminders and alerts
    - Data exports

Persona 4: Retiree Nature Enthusiast ("Mr Tan")

- **Age / Background:** 67, retired, regular park visitor
- **Technical proficiency:** Low
- **Devices:** Budget Android phone

- **Behaviors:**
    - Joins morning walks
    - Struggles with small text and complex forms
- **Needs:**
    - Large, readable UI
    - Very simple workflows
    - Clear error messages and confirmations

Persona 5: Busy Working Professional (“Aisha”)

- **Age / Background:** 29, marketing executive
- **Technical proficiency:** High
- **Devices:** Mobile phone
- **Behaviors:**
    - Signs up last minute
    - Cancels if work overruns
- **Needs:**
    - Fast join/cancel actions
    - Calendar reminders
    - Clear partner status notifications

2.3 Constraints

Platform and Interface Constraints

- The system shall support Google Chrome as the target browser.
- The system interface language shall be English only.

Operational Constraints

- Each survey walk slot shall have a maximum capacity of three (3) volunteers.
- Each observation shall allow the attachment of photo and/or video media, with a maximum of ten (10) media files per observation.

Data Integrity Constraints

- GPS-based location data shall be mandatory for all submitted observations (including "Not Sighted" outcomes).
- Once an observation is submitted, it shall be immutable and not editable by volunteers.

Visibility and Safety Constraints

- Observation details, including location and sighting outcomes, shall not be publicly visible during active survey rounds.
- The system shall prevent real-time or near real-time disclosure of primate locations to protect wildlife from disturbance or misuse.
- During active survey rounds, access to observation data shall be restricted to authorised administrators and volunteers assigned to the relevant slot group.

Regulatory and Ethical Constraints

- The system shall comply with applicable personal data protection regulations (e.g., Singapore's Personal Data Protection Act).
- Personal information of volunteers shall be accessible only to authorised administrators.
- All stored data must be protected against unauthorised access and accidental loss.

2.4 Assumptions and Dependencies

Assumptions

- Volunteers have access to at least one device capable of running Google Chrome (e.g., smartphone, tablet, or computer).
- Volunteers have at least intermittent internet connectivity before or after survey walks in order to synchronise data.
- Survey walk routes, locations, or time slots may change due to weather conditions, park access restrictions, or safety concerns.
- "Not-sighted" outcomes are considered valid and valuable scientific data and must be recorded in the system.
- Volunteers will bring their mobile devices with them during field walks.

Dependencies

- An external email notification service is required to send automated reminders and cancellation alerts.
- A media storage service is required to store uploaded photos and videos associated with observations.
- GPS services provided by user devices and browsers are required to capture location data for observations.
- An authentication component and persistent data storage are required to support secure access control and long-term data retention.
- An external map visualisation service is required to support map visualisation of GPS observations.

3. Requirements Analysis

3.1 Functional Requirements

3.1.1 User Requirements

Account & Participation

- **UR-01:** The system shall allow volunteers to create an account, which must be approved by an administrator before the volunteer may participate in survey rounds.
- **UR-02:** The system shall allow volunteers to view available walk slots and sign up for a slot.
- **UR-03:** The system shall enforce a maximum slot capacity of three (3) volunteers.

Walk Execution & Outcomes

- **UR-04:** The system shall allow volunteers to record walk completion status: Completed, Partially Completed, or Aborted.
- **UR-05:** The system shall allow volunteers to record both sighted and not-sighted outcomes.

Observation Reporting

- **UR-06:** The system shall support observation reporting both online and offline, with later synchronisation when an internet connection becomes available.
- **UR-07:** The system shall require GPS-based location data for each submitted observation, including “Not Sighted” outcomes.
- **UR-08:** The system shall allow volunteers to attach up to ten (10) media files (photos or videos) to an observation.
- **UR-09:** The system shall support draft observations that can be saved and later submitted.
- **UR-10:** The system shall prevent volunteers from editing an observation after submission.

Flexible Data Capture

- **UR-11:** The system shall support approximate primate counts (e.g., ranges or “Unknown”).
- **UR-12:** The system shall allow volunteers to enter free-text behaviour descriptions and notes without requiring predefined categories.

Coordination & Visibility

- **UR-13:** The system shall allow volunteers in the same slot group to view the group’s submitted reports for that slot.
- **UR-14:** The system shall allow volunteers to view their own participation progress and past records.

Guidance

- **UR-15:** The system shall provide lightweight beginner guidance for first-time volunteers.

Notifications

- **UR-16:** The system shall notify volunteers of upcoming walks.
- **UR-17:** The system shall notify volunteers when a slot member cancels.

Administration

- **UR-18:** The system shall allow administrators to open and manage survey rounds.
- **UR-19:** The system shall allow administrators to create, update, and remove walk slots.
- **UR-20:** The system shall allow administrators to bulk-create walk slots.
- **UR-21:** The system shall provide administrators with participation statistics.
- **UR-22:** The system shall provide administrators with indicators for late cancellations and high participation based on administrator-configurable thresholds.

Data Portability

- **UR-23:** The system shall support exporting data in a spreadsheet-compatible format.
- **UR-24:** The system shall support importing legacy data in a spreadsheet-compatible format.

Incident Follow-up

- **UR-25:** The system shall allow volunteers to flag urgent incident types (e.g., traps, feeding, roadkill) for follow-up by administrators.

3.1.2 System Requirements

1. Use Cases Overview

Actors:

- Volunteer
- Administrator

Volunteer use cases:

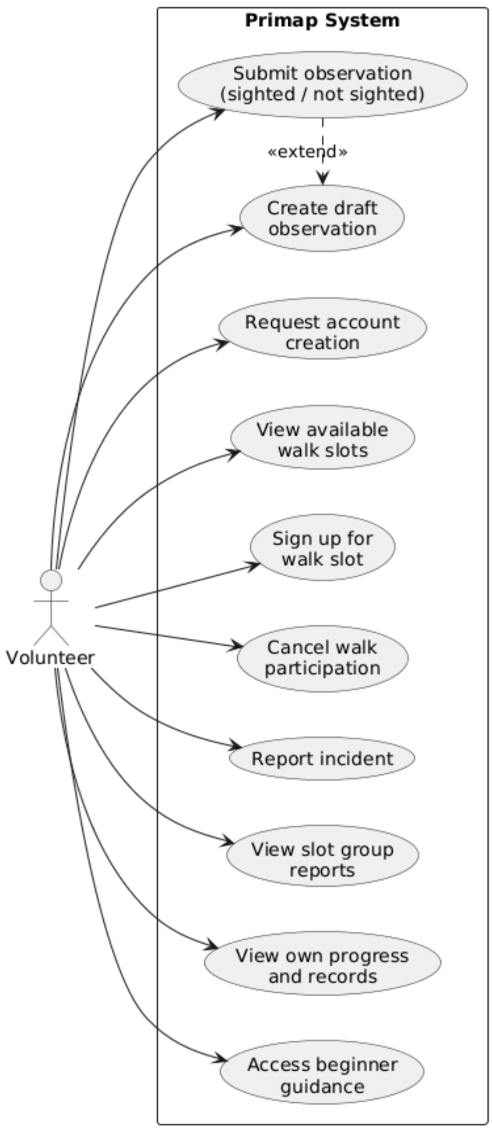
- Request account creation
- View available walk slots
- Sign up for a walk slot
- Cancel walk participation
- Create draft observation
- Submit observation
- Upload media
- Record walk completion status

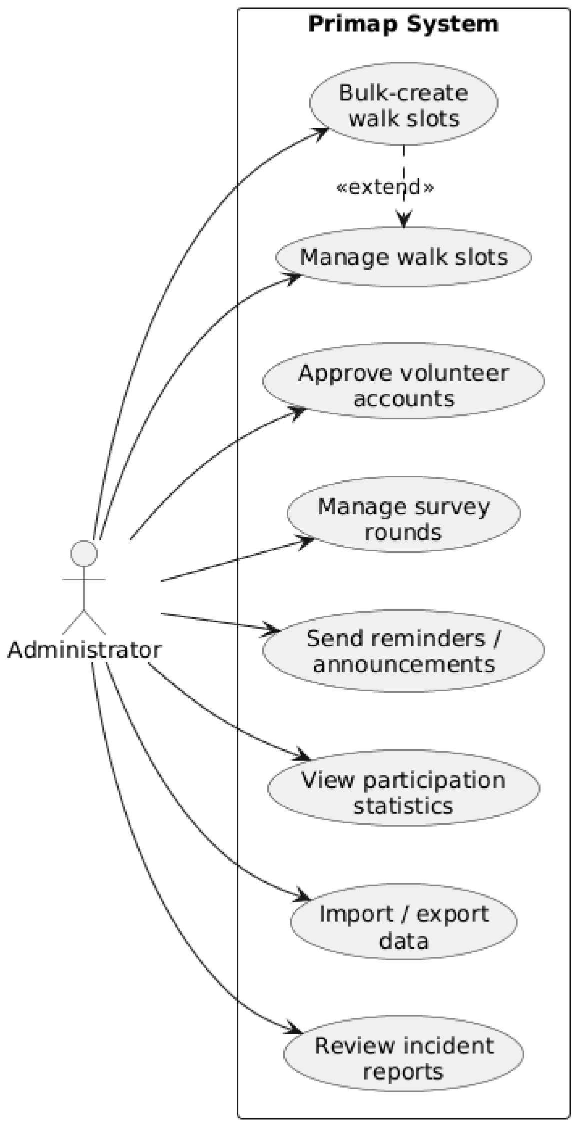
- View slot group reports
- View own progress and records
- Access beginner guidance
- Receive reminders and cancellation notifications
- Report incident

Admin use cases:

- Approve volunteer account
- Manage survey rounds
- Manage walk slots (CRUD)
- Bulk-create walk slots
- Send reminders/announcements
- View participation statistics and indicators
- Import data
- Export data

The following use case diagrams summarise these interactions.





2. Important Use Cases

UC-01 Request Account Creation

Actors: Volunteer

Related URs: UR-01

Description: A volunteer creates an account request in Primap. The account remains inactive until approved by an administrator.

Preconditions:

- Volunteer is not logged in.
- Volunteer does not already have an active account.

Basic Flow:

1. Volunteer opens Primap in their own browser.
2. Volunteer selects “Create account”.
3. System prompts for required registration information.
4. Volunteer submits registration information.
5. System validates the submitted information.
6. System creates the account in a Pending Approval state.
7. System informs the volunteer that administrator approval is required before participation.

Alternative Flows:

- A1 (Invalid input): System highlights invalid/missing fields; volunteer corrects and resubmits.
- A2 (Email already exists): System informs volunteer and directs to login/support flow.

Postconditions:

- A new volunteer account exists in a Pending Approval state.
- Volunteer cannot sign up for survey walks or submit observations until approved.

UC-02 View Available Walk Slots

Actors: Volunteer

Related URs: UR-02

Description: Volunteer views upcoming walk slots open for sign-up.

Preconditions:

- Volunteer is logged in.
- At least one survey round exists.

Basic Flow:

1. Volunteer navigates to “Walk Slots”.
2. System displays available slots with date, time window, location, and remaining capacity.

Alternative Flows:

- A1 (No available slots): System displays “No available slots”.

Postconditions:

- Available walk slots are displayed.

UC-03 Sign Up for a Walk Slot

Actors: Volunteer

Related URs: UR-02, UR-03

Description: Volunteer signs up for an available walk slot.

Preconditions:

- Volunteer is logged in and approved.
- The selected slot exists.

Basic Flow:

1. Volunteer selects a walk slot.
2. System checks that the slot is not full (capacity < 3).
3. System displays sign-up confirmation.
4. Volunteer confirms sign-up.
5. System registers the volunteer into the slot group.
6. System confirms successful sign-up.

Alternative Flows:

- A1 (Slot full): System blocks sign-up.
- A2 (Already signed up): System displays a warning and does not duplicate sign-up.

Postconditions:

- Volunteer is added to the slot group.

UC-04 Cancel Walk Slot Participation

Actors: Volunteer

Related URs: UR-02, UR-17

Description: Volunteer cancels participation in a previously signed-up walk slot.

Preconditions:

- Volunteer is logged in.
- Volunteer is signed up for the selected walk slot.

Basic Flow:

1. System displays the volunteer's upcoming walk slots.
2. Volunteer selects "Cancel" for a slot.
3. System requests confirmation.
4. Volunteer confirms cancellation.
5. System removes the volunteer from the slot.
6. System notifies remaining slot members of the cancellation.

Alternative Flows:

- A1 (Slot not found / already cancelled): System informs the volunteer and refreshes the view.

Postconditions:

- Volunteer is removed from the slot.
- Remaining slot members are notified.

UC-05 Create Draft Observation

Actors: Volunteer

Related URs: UR-06, UR-09, UR-12

Description: Volunteer creates a draft observation for a walk slot, including optional free-text notes and media, even if offline.

Preconditions:

- Volunteer is logged in and assigned to the walk slot.

Basic Flow:

1. Volunteer selects an assigned walk slot.
2. System displays the slot's group reports and observation list.
3. Volunteer selects "Add Observation".
4. System displays the observation draft form.
5. Volunteer enters available information (including free-text notes) and may attach media.
6. System saves the observation as a draft (offline or online).

Alternative Flows:

- A1 (Offline): System saves the draft locally and marks it "Pending sync".

- A2 (Optional GPS metadata detected in media): System may propose extracted GPS for volunteer confirmation.

Postconditions:

- A draft observation exists and can be edited prior to submission.

UC-06 Submit Observation

Actors: Volunteer

Related URs: UR-04, UR-05, UR-06, UR-07, UR-10, UR-11, UR-12

Description: Volunteer submits an observation for a walk slot. Submission records completion status and locks the report.

Preconditions:

- Volunteer is logged in and assigned to the walk slot.
- A draft observation exists or volunteer is entering a new submission.

Basic Flow:

1. Volunteer selects the walk slot and selects “Submit observation”.
2. System requests walk completion status (Completed/Partially Completed/Aborted).
3. Volunteer selects completion status and enters any remarks (optional).
4. System requests outcome (Sighted/Not Sighted).
5. If Sighted, system requests primate species and count (including approximate formats).
6. System captures GPS location (or prompts for manual pinning if needed).
7. System requests final confirmation.
8. Volunteer confirms submission.
9. The system submits the observation and locks it from editing.

Alternative Flows:

- A1 (GPS unavailable): System blocks submission and prompts retry or manual pin.
- A2 (Required fields missing): System blocks submission and highlights missing fields.
- A3 (Offline at submission): System saves as draft and queues submission until online.
- A4 (Already submitted): System blocks and displays “Observation already submitted”.

Postconditions:

- Observation is submitted and immutable.

UC-07 Upload Media

Actors: Volunteer

Related URs: UR-08

Description: Volunteer attaches media to an observation draft (up to 10 files).

Preconditions:

- Volunteer is logged in.
- A draft observation exists.

Basic Flow:

1. Volunteer opens the draft observation form.
2. Volunteer selects media files to upload (max 10).
3. System attaches the files to the draft.
4. Volunteer saves the draft.

Alternative Flows:

- A1 (More than 10 files): System blocks and requests reducing the selection.

Postconditions:

- Media is associated with the draft observation.

UC-08 View Slot Group Reports

Actors: Volunteer

Related URs: UR-13

Description: Volunteer views observations submitted by members of the same slot group.

Preconditions:

- Volunteer is logged in and assigned to the slot.

Basic Flow:

1. Volunteer selects an assigned walk slot (past or ongoing).
2. System displays group observation submissions and statuses.

Postconditions:

- Slot group reports are displayed.

UC-09 View Own Progress and Records

Actors: Volunteer

Related URs: UR-14

Description: Volunteer views past walk participation and progress toward required minimum walks (if applicable).

Preconditions:

- Volunteer is logged in.

Basic Flow:

1. Volunteer navigates to “My Progress”.
2. System displays:
    - number of signed-up slots,
    - remaining required walks (if applicable),
    - list of past slots and their outcomes.

Postconditions:

- Participation progress and records are displayed.

UC-10 Access Beginner Guidance

Actors: Volunteer

Related URs: UR-15

Description: Volunteer accesses lightweight beginner guidance (tips for first-timers).

Preconditions:

- Volunteer is logged in.

Basic Flow:

1. Volunteer navigates to “Guidance”.
2. System displays beginner guidance content.

Postconditions:

- Guidance content is displayed.

UC-11 Report Incident

Actors: Volunteer

Related URs: UR-08, UR-25

Description: Volunteer reports an urgent incident (e.g., traps, feeding, roadkill) for administrator follow-up.

Preconditions:

- Volunteer is logged in and assigned to a walk slot.

Basic Flow:

1. Volunteer opens an assigned walk slot.
2. Volunteer selects “Report Incident”.
3. System displays incident form with incident type choices.
4. Volunteer selects an incident type and enters details (optional).
5. Volunteer may attach media (max 10).
6. Volunteer submits the incident report.
7. System notifies administrators that an incident report was submitted.

Postconditions:

- Incident report is stored and visible to administrators.

UC-12 Bulk Create Walk Slots

Actors: Administrator

Related URs: UR-18, UR-19, UR-20

Description: Administrator bulk-creates walk slots based on scheduling rules.

Preconditions:

- Administrator is logged in.

Basic Flow:

1. Administrator navigates to “Bulk Create Walk Slots”.
2. System displays bulk creation form.
3. Administrator enters bulk rules (day/time range, start/end week, etc.).
4. System previews generated slots.
5. Administrator confirms creation.
6. System creates the walk slots.

Postconditions:

- Walk slots are created based on schedule set.

UC-13 Approve Volunteer Account

Actors: Administrator

Related URs: UR-01

Description: Administrator reviews a pending account request and approves it to allow participation.

Preconditions:

- Administrator is logged in.
- At least one account exists in a Pending Approval state.

Basic Flow:

1. Administrator navigates to “Pending Accounts”.
2. System displays pending account requests.
3. Administrator selects a request.
4. System displays request details.
5. Administrator selects “Approve”.
6. System updates the account to Active.
7. System confirms approval to the administrator.
8. System notifies the volunteer of approval.

Alternative Flows:

- A1 (Reject request): System marks as Rejected; volunteer cannot participate.
- A2 (Request already processed): System informs admin and refreshes list.

Postconditions:

- Account is Active (approved) or Rejected.

UC-14 View Participation Statistics and Indicators

Actors: Administrator

Related URs: UR-21, UR-22

Description: Administrator views participation statistics and indicators such as late cancellations and high participation.

Preconditions:

- Administrator is logged in.

Basic Flow:

1. Administrator navigates to “Statistics”.
2. System displays participation summaries (e.g., sign-ups, completions, cancellations).
3. System displays indicators based on configured thresholds.

Postconditions:

- Statistics and indicators are displayed.

UC-15 Configure Indicator Thresholds

Actors: Administrator

Related URs: UR-22

Description: Administrator configures thresholds used to classify late cancellations and high participation.

Preconditions:

- Administrator is logged in.

Basic Flow:

1. Administrator opens “Settings”.
2. Administrator edits indicator thresholds.
3. Administrator saves settings.
4. System applies the updated thresholds.

Postconditions:

- Threshold settings are updated.

UC-16 Import Legacy Data

Actors: Administrator

Related URs: UR-24

Description: Administrator imports legacy spreadsheet-format data into Primap.

Preconditions:

- Administrator is logged in.
- Import file is available.

Basic Flow:

1. Administrator selects “Import Data”.
2. System prompts for a compatible spreadsheet file.
3. Administrator uploads the file.
4. System validates format and fields.
5. System imports data and reports summary results.

Alternative Flows:

- A1 (Invalid format): System rejects file and reports errors.

Postconditions:

- Imported data is stored in Primap.

UC-17 Export Data

Actors: Administrator

Related URs: UR-23

Description: Administrator exports data in a spreadsheet-compatible format.

Preconditions:

- Administrator is logged in.

Basic Flow:

1. Administrator selects “Export Data”.
2. Administrator selects scope (round, date range, all data).
3. System generates export file.
4. System provides export file to administrator.

Postconditions:

- Export is generated successfully.

3.2 Non-Functional Requirements

Usability

- **Requirement:** At least 80% of first-time volunteers shall be able to (a) sign up for a walk slot and (b) submit a report within 10 minutes, without external help.
- **Rationale:** Volunteers vary widely in technical proficiency and the ease of use of the app is an important goal of some stakeholders as it would affect the quality of the volunteering experience
- **Test:** Test with first-time users and measure completion rate and time.

Reliability

- **Requirement:** When offline, the system shall allow creation and saving of draft observations and shall synchronise drafts without data loss once connectivity is restored, in ≥ 99% of tested sync scenarios.
- **Rationale:** Survey routes may have poor coverage and reporting later is common so the system should be able to handle offline scenarios reliably to prevent loss of valuable sighting data and giving volunteers a poor user experience
- **Test:** Automated sync test cases across simulated offline/online transitions.

Performance

- **Requirement:** For 95% of requests, the system shall load the “Available Walk Slots” list within 3 seconds on a typical mobile network and within 2 seconds on desktop broadband.
- **Rationale:** A good user experience for volunteers is important to encourage participation and improve their overall volunteering experience. An overly slow load time might discourage potential volunteers from participating actively
- **Test:** Performance testing with network throttling

Privacy & Ethics

- **Requirement:** During an active round, observation details shall be accessible only to (a) administrators and (b) volunteers assigned to the relevant slot group; the system shall not provide public sharing endpoints.
- **Rationale:** One of the requirements is to not broadcast sightings during survey periods so as to prevent public individuals from disrupting the Langurs’ natural habitats.
- **Test:** Access control tests for unauthorised users and non-slot volunteers.

3.3 Data Requirements

The system shall store, at minimum, all data currently captured in the legacy spreadsheets. This includes:

Volunteer:

- name *(non-empty string)*
- email *(valid email, unique)*
- role *(Volunteer/Admin)*
- status *(Pending Approval/Active/Disabled/Rejected)*

Survey Round:

- start date *(YYYY-MM-DD)*
- end date *(YYYY-MM-DD)*
- status *(Open/Closed)*

Walk Slot:

- date *(YYYY-MM-DD)*
- time window *(HH:MM–HH:MM)*
- location descriptor *(optional free text, e.g., "Lower Pierce boardwalk")*
- capacity *(integer ≤ 3)*
- assigned volunteers *(0–3, with per-volunteer participation status: Active/Cancelled)*
- completion status *(Completed/Partially Completed/Aborted/Unrecorded)*

Observation:

- associated walk slot reference *(mandatory)*
- date/time *(YYYY-MM-DD, HH:MM or time range or NA)*
- outcome *(Sighted/Not Sighted)*
- location *(latitude/longitude decimal degrees; mandatory on submission)*
- primate species *(RBL/LTM/Dusky; required if Sighted)*
- count *(integer or range "x–y" or Unknown)*
- behaviour/notes *(free text)*

Media:

- type *(Photo/Video)*
- associated observation *(mandatory)*
- reference/identifier *(stored reference to retrieve media)*

Incident Flag:

- type *(Trap/Feeding/Roadkill/Other)*
- associated walk slot reference *(mandatory)*
- details *(free text)*